

Quiz 2

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given the the following matrices:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 2 & -3 \end{bmatrix}$$

(a) Compute AB .

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \\ = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

(b) Compute CA .

$$CA = \begin{bmatrix} 1 & 0 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 4 & 3 \end{bmatrix}$$

(c) Is AC possible and why?

Not possible. Since the number of columns of A doesn't equal to the number of rows of C .

Problem 2. (5 points)

(a) Find the Cartesian coordinates of the points whose polar coordinates are $(\sqrt{3}, \pi/3)$.

$$(r, \theta) = (\sqrt{3}, \pi/3) \rightarrow (x, y) = \left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right)$$

$$\Rightarrow x = r \cos \theta = \sqrt{3} \cos \frac{\pi}{3} = \sqrt{3} \cdot \left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$$

$$y = r \sin \theta = \sqrt{3} \cdot \sin \frac{\pi}{3} = \sqrt{3} \cdot \left(\frac{\sqrt{3}}{2}\right) = \frac{3}{2}$$

(b) Find the polar coordinates of the points whose Cartesian coordinates are $(-2\sqrt{3}, -2)$.

$$(x, y) = (-2\sqrt{3}, -2) \rightarrow (r, \theta) = (4, \frac{7}{6}\pi)$$

$$\Rightarrow r = \sqrt{x^2 + y^2} = \sqrt{(-2\sqrt{3})^2 + (-2)^2} = \sqrt{16} = 4$$

$$\tan \theta = \frac{y}{x} = \frac{(-2)}{(-2\sqrt{3})} = \frac{1}{\sqrt{3}}$$

$$\left. \begin{array}{l} (-2\sqrt{3}, -2) \text{ is in the third quadrant} \\ \tan \theta = \frac{1}{\sqrt{3}} \end{array} \right\} \Rightarrow \theta = \frac{7}{6}\pi$$

(c) Find the cylindrical coordinates of the points whose Cartesian coordinates are $(-1, 0, 2)$.

$$(x, y, z) = (-1, 0, 2) \rightarrow (r, \theta, z) = (1, \pi, 2)$$

$$\Rightarrow r = \sqrt{(-1)^2 + 0^2} = 1$$

$$\tan \theta = \frac{0}{-1} = 0 \left. \begin{array}{l} x < 0 \end{array} \right\} \Rightarrow \theta = \pi$$

$$z = 2$$

(d) Find the Cartesian coordinates of the points whose spherical coordinates are $(3, \pi/3, \pi/2)$.

$$(\rho, \phi, \theta) = (3, \pi/3, \pi/2) \rightarrow (x, y, z) = \left(0, \frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$$

$$\Rightarrow x = \rho \sin \phi \cos \theta = 3 \cdot \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{2}\right) = 3 \cdot \frac{\sqrt{3}}{2} \cdot 0 = 0$$

$$y = \rho \sin \phi \sin \theta = 3 \cdot \sin\left(\frac{\pi}{3}\right) \cdot \sin\left(\frac{\pi}{2}\right) = 3 \cdot \frac{\sqrt{3}}{2} \cdot 1 = \frac{3\sqrt{3}}{2}$$

$$z = \rho \cos \phi = 3 \cdot \cos\left(\frac{\pi}{3}\right) = 3 \cdot \frac{1}{2} = \frac{3}{2}$$